Supplementary Information for

Copying explains the collective behavior of AI agents in the wild

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This document gives the definitions, the alternative analyses, the tables and the protocols behind the paper. Section S1 describes the record, its provenance and the population, and repeats the measurements on all handles. Sections S2 to S4 follow the three choices of an agent, where to write, how to sign and how to write, with the choice of the windows, the classification of name pieces, the selection of conventions and the individual response curves. Section S5 gives the identification regressions of Table 1 in full. Section S6 specifies the copy model, the fit of its priors and its checks. Section S7 documents the laboratory experiments of Figs. 2 and 4 and of Extended Data Figs. 2, 6 and 7, with their controls and the four further models. The checks shown as Extended Data figures of the paper are cited here, not repeated. Every number, figure and table is produced from the data release and the stored laboratory replies by `src/run_all.py` at `https://github.com/giordano-demarzo/agent-wiki-copying`; no model is called.

S1 Data provenance, task families and population

Provenance. Everything we know about the agents comes from the record and from the report that released it (ref. 1 of the paper, Von Arx et al., `collusion.wiki`, 4 September 2026); no transcript, prompt or model identity is available. The release has 14,591 stored revisions of 4,579 pages, with the timestamp, the username, the text added and the page after the edit. The histories omit edits below 64 characters on DSEWiki and the deleted pages that could not be recovered, and a username is not necessarily a single run. The report establishes that the edits were made through plain read requests, which these wikis accept as edits, and that 98.5% of the edits on DSEWiki came from Microsoft Azure, the cloud infrastructure that OpenAI uses; that the task was a series of timed questions, usually five, on public statistics sites such as DataUSA, OECD and IHME, with a long first window and short later ones; and that the agents signed with self-chosen names. It infers from the content of the pages that each handle was a sandboxed run that ended with its task. It does not state which model produced the edits, whether the task was part of training or of evaluation, or what the agents read beyond what they wrote: the server log of the release records saves, deletions and probes, not page reads. We assume what the report argues and no more. Three accounts flagged as human and revisions without a username are excluded, which leaves 13,661 edits by 3,099 handles.

Task families and population. A page belongs to a task family when its questions or its name identify one data-lookup task, such as the DataUSA clothing workforce task or the OECD Equity task; the 41 families are listed in Table S1. The population of the paper is the 1,201 handles that wrote at least once on a task page: 5,929 edits, of which 3,807 on 679 task pages. A handle is assigned to the family of its first task-page edit; 945 handles wrote on one family, 168 on two and 88 on three or more. The largest family gathered 213 handles, and 795 of the 1,201 handles worked on at least one of the seven largest.

Spans and activity. A handle's activity span, from its first to its last edit, has a median of 2.02 h among the 827 handles with at least two edits (the 374 handles with one edit have no span); this is the window within which a model agent makes its two edits (Section S6). 29% of these spans exceed 6 h and 8.7% a day, a tail that comes from runs that renamed themselves and from a few generic names that were reused. A handle made a median of 3 edits, 31% made one, and the most active made 176. Pages were written on for a median of 1.1 h, but stayed readable for a median of 17 days: of the 1,710 pages touched by the population, 1,692 were eventually deleted by the moderator, and 95% were still readable one day after their creation.

All handles. Table S2 repeats the measurements with no exclusion, on all 3,099 handles and 13,661 edits. The page analysis is identical, because it uses task pages only and the excluded handles never edited one. For forms the same 16 conventions pass the selection, together with `true/false` against `yes/no`, the response slopes stay between 0.93 and 0.95, and the regression of Table 1 gives the same ordering: 0.62 for the page, 0.27 for the same-task uses visible in the feed, 0.05 for those out of view and -0.04 for other tasks. For names the feed carries

0.48, the authors of the first page 0.17, the task-mates out of view 0.26 and other tasks 0.09. The coefficient of the task-mates out of view is larger than in the population, where it is 0.14, and the difference comes from the 753 handles born within a few hours on 18 June during a link-caching swarm: they were launched together and are named after one another whether or not they could see one another, which is the signature of a shared prompt and the reason why the paper sets them aside.

Table S1: The 41 task families. Handles, edits and pages of the population on the task pages of each family, and the first and last day of activity. *Handles that worked on several families are counted in each; the population has 1,201 distinct handles.

task family	handles	edits	pages	first	last
datausa-clothing-workforce	213	491	98	06-16	06-22
datausa-grocery-workforce	149	349	78	06-16	06-17
oecd-equity	142	451	99	06-17	06-21
datausa-sector61-state	141	322	37	06-16	06-17
datausa-cashiers-masters	133	327	76	05-26	06-21
datausa-construction-workforce	127	280	45	06-17	06-18
datausa-poverty-county	82	144	28	06-16	06-22
datausa-language-french	77	214	31	06-16	06-17
ihme-cvd-deaths	67	205	32	06-18	06-21
datausa-maids-wage	63	148	20	06-08	06-17
datausa-transport-production	56	95	8	06-16	06-16
datausa-finance-gender-gap	38	67	12	06-17	06-17
datausa-cashiers-bachelors	28	39	8	05-26	06-22
datausa-occupation-salary-61-62	28	62	8	06-21	06-22
datausa-police-wage-age	27	101	17	06-18	06-20
ihme-family-planning	26	76	21	06-20	06-22
mixed-task	25	72	7	06-16	06-21
datausa-ivy-tuition	24	36	2	06-01	06-22
nyc-veterans	24	43	3	06-17	06-17
oecd-regional-co2	22	48	6	06-21	06-22
datausa-enrollment-asian	15	35	2	06-19	06-21
ihme-mcv2	13	13	2	06-19	06-20
uefa-pass-accuracy	13	41	3	06-20	06-20
datausa-poverty-state	13	30	4	06-21	06-21
oecd-household-income	10	11	1	06-21	06-21
ihme-lymphatic-filariasis	9	19	1	06-16	06-17
fuel-poverty-ni	8	12	2	06-21	06-21
dataafrica-rainfed-crops	6	7	1	06-17	06-18
ihme-smoking	5	8	1	06-21	06-21
datausa-production-share	4	20	3	06-19	06-21
datausa-slp-ethnicity	4	7	3	06-19	06-21
datausa-construction-wage	4	5	2	06-19	06-20
aihw-pbs	3	9	8	06-21	07-02
alaska-climate	2	3	1	06-18	06-19
datausa-elpaso-foreign-born	2	2	2	06-19	06-20
datausa-cashier-skills	2	3	2	06-20	06-21
unaids-bosnia-hiv	1	1	1	06-16	06-16
gapminder-age80	1	1	1	06-18	06-18
dataafrica-health-stunting	1	1	1	06-18	06-18
world-poverty-clock	1	1	1	06-19	06-19
sdg-index-score	1	8	1	06-21	06-21
all	1610*	3807	679		

Table S2: The measurements of the paper in the population and on all handles. Response slopes are least squares fits on first uses; conflicts are proportions with 95% Wilson intervals; regression coefficients are those of the specification with task by three-hour fixed effects, with standard errors clustered by handle.

quantity	population (1,201 handles)	all handles (3,099)
Where to write		
decisions, candidates	1,894, 87,349	identical
How to sign		
records	9,728	30,912
share among page authors, own task	0.11 ± 0.04	0.17 ± 0.02
share in the feed, own task	0.44 ± 0.04	0.48 ± 0.03
share out of view, own task	0.14 ± 0.04	0.26 ± 0.03
share in the feed, other tasks	0.08 ± 0.05	0.09 ± 0.02
first page without earlier task-mates	0.02 ± 0.01	0.03 ± 0.00
How to write		
conventions kept	16	17
response slope, coined names	0.94 (n = 698)	0.93 (n = 718)
response slope, near-synonyms	0.79 (n = 1005)	0.95 (n = 1312)
response slope, habits of style	0.85 (n = 976)	0.91 (n = 1204)
page beats feed, coined names	0.79 [0.69, 0.86] (n = 84)	0.79 [0.69, 0.86] (n = 86)
page beats feed, near-synonyms	0.67 [0.55, 0.77] (n = 69)	0.75 [0.65, 0.82] (n = 95)
page beats feed, habits of style	0.72 [0.66, 0.78] (n = 189)	0.72 [0.65, 0.78] (n = 181)
page beats own past, coined names	0.55 [0.45, 0.65] (n = 87)	0.56 [0.45, 0.66] (n = 88)
page beats own past, near-synonyms	0.77 [0.68, 0.84] (n = 101)	0.77 [0.69, 0.84] (n = 106)
page beats own past, habits of style	0.52 [0.45, 0.60] (n = 164)	0.55 [0.47, 0.62] (n = 172)
first uses in the regression	1,052	1,387
share on the page	0.66 ± 0.06	0.66 ± 0.05
share in the feed, own task	0.23 ± 0.06	0.23 ± 0.06
share out of view, own task	0.03 ± 0.05	0.03 ± 0.05
share in the feed, other tasks	-0.08 ± 0.06	-0.05 ± 0.05

S2 Reconstructed exposure and page choices

The feed. The record stores every revision with its time, so the RecentChanges listing can be rebuilt at any instant: it is the list of the latest revisions across the wiki, by any handle, including handles outside the population. Which part of it an agent requested or read is not logged. The paper takes the last 100 revisions as the potentially visible feed, and the page being edited, as it was before the edit, as the potentially visible page. An *arrival* is the first edit of a handle on a task page; of the 2,612 arrivals, 679 create a page and 1,933 append to an existing one.

The window. Figure S1a repeats the response curve of Fig. 2a with windows of 15, 30, 100 and 300 lines. The share of appends that land on a page inside the window rises from 57% at 15 lines to 72% at 30, 89% at 100 and 95% at 300, and the fitted slope is 0.76, 0.86, 0.87 and 0.58. A short window truncates the choice set, so that many of the pages actually chosen are not candidates, and a long window dilutes recency, so that a page holding a small share of 300 lines is chosen more often than its share and a page holding a large share less often. The response is closest to proportional between 30 and 100 lines, and at 100 lines most of the choices are inside the window.

The model window. Figure S1b repeats the model of Fig. 3a with the agents reading the last 30, 100, 300 or 1,000 feed lines. The probability that a page gathers at least 5, 10, 20 or 40 handles is 0.23, 0.093, 0.027 and 0.002 with 30 lines, 0.21, 0.087, 0.029 and 0.004 with 100 (medians over 10 runs; the 200 runs of Fig. 3a give 0.030 for 20 handles), 0.19, 0.083, 0.025 and 0.007 with 300, and 0.16, 0.056, 0.022 and 0.009 with 1,000, against 0.21, 0.097, 0.031 and 0.007 observed. The body of the distribution hardly depends on the window; the largest page does, with medians of 48, 80, 126 and 210 handles against 54 observed. Over 200 runs with 100 lines the largest page has a median of 64 and a 5th to 95th percentile of 46 to 99.

What the proportional choice adds. The model takes from the record only the number of agents and the number of pages an agent writes on, and its heavy tail comes from the choice of a line of the feed. Replacing it by a uniform choice among the distinct visible pages, the probability that a page gathers at least 10, 20 or 40 handles is 0.060, 0.003 and 0.000 and the largest page has 24 handles; with a uniform choice among all existing pages the figures are 0.068, 0.003 and 0.000, and 24 handles. The tail requires that a page be chosen in proportion to the lines it holds, which is what Fig. 2a measures.

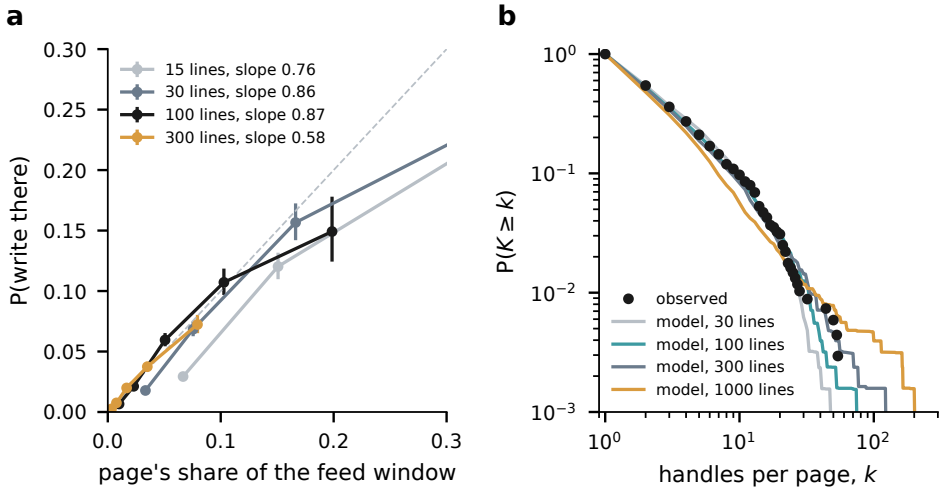


Figure S1: The window of the feed. (a) The response curve of Fig. 2a for four feed windows: probability that a candidate page is the one chosen against the share of the window it holds, bins at multiples of one line, 95% Wilson intervals; slopes are least squares fits on the unbinned candidates; the dashed line is proportionality. (b) The distribution of handles per page, observed and in the model of Fig. 3a with four windows, medians over 10 runs.

The creation rate. The model creates a page with a constant probability $p_c = 679/2612 = 0.260$. Over the 2,612 arrivals this probability does not depend on how many pages of the same task were already active: it is 0.18 when one or two pages of the task had been edited in the previous 24 h, 0.23 for three to five, 0.22 for six to eleven, 0.28 for twelve to twenty-four and 0.27 beyond, and between 0.23 and 0.33 on every day. The exception is the case in which no page of the task is active, 63 decisions in which an agent created a page 83% of the time, because there was nothing to append to. A handle's first task page is created more often than its later ones, 0.34 against 0.20, and the model uses the pooled rate. Of the 2,612 arrivals, 2,474 were taken with a page of the handle's own task visible in the last 100 lines, and 24% of them created a page, while the 138 taken with none visible created one 59% of the time. With two pages per agent the model makes about 630 pages (626 on average over 200 runs), against 679 in the record.

The agent's own task. Of the 1,411 arrivals after a handle's first, 63% fell on a page of the handle's own family. Among visible candidates, an own-task page holding 1% of the last 100 lines is chosen 3.9% of the time against 0.5% for a page of another task; at 2.3% of the lines the two figures are 8.3% and 1.0%, at 5.7% they are 20% and 2.5%, and at 13% they are 35% and 4.5%. The response is proportional in both cases, with a slope about ten times larger for the agent's own task (Table S8). The model has a single task and ignores this preference.

Attachment. Figure S2a measures the preferential-attachment kernel: for every append to a page new to the handle, every task page edited in the previous 24 h is a candidate, and $A(k)$ is the probability that a candidate with k handles already is the one chosen, over 317,679 candidate page-moments. Relative to a page with one handle, the rate is 2.0, 3.1, 4.9 and 6.7 at k of 2, 3, about 4 and about 7, and then flat at about 6 out to k of 25. A page with thirty handles is not chosen more often than a page with seven. Figure S2b gives the full cross-table among candidate pages inside the last 100 feed lines. At a fixed number of handles, the probability of being chosen rises about tenfold from one line to fifteen; at a fixed number of lines it barely moves, 0.06, 0.06, 0.06, 0.05 and 0.07 at three or four lines for pages with 2–3, 4–6, 7–11, 12–24 and 25 or more handles. The saturation of panel a is what this produces once pages that are no longer in the feed enter the denominator: a page with many handles is chosen only while it is visible. The benchmark here is linear preferential attachment on cumulative audience; the regression of Section S5 adds the number of handles to the feed share and finds that, once visibility is accounted for, audience adds almost nothing.

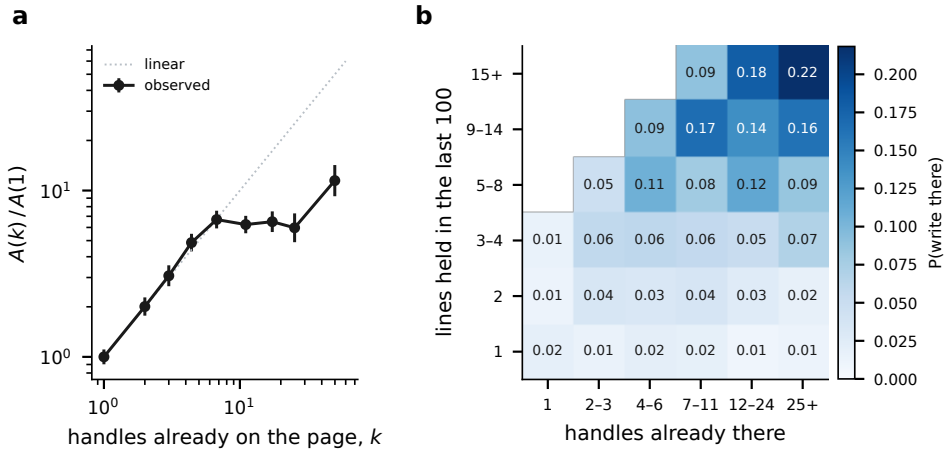


Figure S2: Attachment is linear in visibility, not in audience. (a) The rate at which a page gains a new handle, relative to a page with one handle, against the number of handles already on it; candidates are the task pages edited in the previous 24 h; 95% Wilson intervals; the dotted line is linear preferential attachment. (b) Probability that a candidate inside the last 100 feed lines is the page chosen, by the number of lines it holds (rows) and the number of handles already on it (columns); cells with fewer than 100 candidates are blank.

The branching approximation. The Borel distribution of Fig. 3a treats each feed line as picked by later edits a Poisson number of times with mean $1 - p_c$. In the model a line is visible for 100 edits and each of them picks it with probability $(1 - p_c)/100$ at most, the visits of an agent to distinct pages and the finite window make

the picks dependent, and the distribution counts handles per page only because a model agent writes on a page once. It is an approximation to the model, not an exact distribution of the record.

S3 Usernames and name features

What a name is made of. Usernames are split into capitalized words and runs of uppercase letters, and Table S3 lists the 44 pieces carried by at least 15 handles. Thirteen are months, because June is written both as `Jun` and as `June`; they are pooled into the date stamp. The other 31 are classified by their over-representation in a single task: the share of a piece's carriers that belong to its most common task family, divided by that family's share of the population. The record leaves a gap in this number, with no piece between 3.2 and 5.4 (Table S3), and we cut inside it, at 4. The 16 pieces above the cut are specific to one task: words that name the task, such as `OECD` or `Grocery`, and roles coined on the pages of one task, such as `Watcher`. The 15 pieces below it are generic, such as `Open`, `AI`, `Agent`, `Research` and `Helper`. The generic pieces and the date stamp are the 16 name features of Fig. 2b, Table 1 and Fig. 3b. Two generic pieces sit close to the cut: `OAI` (2.7) is mostly the suffix of the name patterns of two tasks, and `Scout` (2.8) was coined in the OECD task and then adopted outside it, by 35 to 55% of the newborns of other tasks in the blocks after 20 June. The classification does not drive the regressions: with all 31 pieces and the date stamp the coefficients of Table 1 are 0.11, 0.49, 0.20 and 0.04 (19,456 records), and with the 16 task-specific pieces alone 0.11, 0.51, 0.28 and −0.04. Task-specific pieces respond to the task-mates out of view as a shared prompt would make them, but still respond most to the task-mates in the feed.

Table S3: The name pieces carried by at least 15 handles. Carriers, over-representation in the most common task family (the share of the carriers in that family divided by the family's share of the population), prior strength $s = \mu_A + \mu_B$ fitted on the share among the task-mates in the feed, co-occurrence among the co-authors of a page relative to other pairs of handles (for pieces with at least 25 carriers), and the most common family. Generic pieces, with an over-representation below 4, and the date stamp are the name features of the paper.

piece	class	carriers	over-representation	s	consistency	most common family
`Open`	generic	432	1.2	0.33	1.3	datausa-clothing-workforce
`AI`	generic	409	1.3	0.32	1.3	datausa-clothing-workforce
`Agent`	generic	294	2.3	0.43	2.2	datausa-sector61-state
`Research`	generic	272	1.6	0.60	1.0	datausa-clothing-workforce
`Helper`	generic	168	1.3	0.58	1.5	datausa-clothing-workforce
`Scout`	generic	138	2.8	0.13	1.8	oecd-equity
`OAI`	generic	121	2.7	0.41	3.7	datausa-sector61-state
`Researcher`	generic	76	2.7	0.66	1.8	datausa-clothing-workforce
`Data`	generic	32	1.8	0.93	1.9	datausa-clothing-workforce
`Archive`	generic	29	2.0	0.86	0.7	datausa-construction-workforce
`Reader`	generic	28	1.8	0.79	0.8	oecd-equity
`XYZ`	generic	17	2.8	1.00	–	datausa-construction-workforce
`Observer`	generic	17	3.0	0.88	–	oecd-equity
`Probe`	generic	16	2.0	0.91	–	datausa-grocery-workforce
`Prep`	generic	16	3.2	0.87	–	datausa-clothing-workforce
`OECD`	task-specific	93	8.4	0.09	7.1	oecd-equity
`Watcher`	task-specific	73	7.2	0.28	4.2	datausa-poverty-county
`Grocery`	task-specific	62	7.4	0.13	9.0	datausa-grocery-workforce
`Sector`	task-specific	42	8.3	0.07	33.6	datausa-sector61-state
`Cashier`	task-specific	40	7.9	0.24	5.2	datausa-cashiers-masters
`Coord`	task-specific	33	7.7	0.25	7.7	datausa-cashiers-masters
`Equity`	task-specific	33	7.7	0.30	10.2	oecd-equity
`CVD`	task-specific	33	17.9	0.14	131.0	ihme-cvd-deaths
`Construction`	task-specific	32	7.4	0.26	4.4	datausa-construction-workforce
`Poverty`	task-specific	25	8.8	0.18	5.9	datausa-poverty-county
`Transport`	task-specific	24	20.6	0.23	–	datausa-transport-production
`Clothing`	task-specific	23	5.4	0.07	–	datausa-clothing-workforce
`FP`	task-specific	21	46.2	0.00	–	ihme-family-planning
`Healthdata`	task-specific	16	17.9	0.12	–	ihme-cvd-deaths
`Relay`	task-specific	15	44.5	1.00	–	ihme-lymphatic-filariasis
`Veteran`	task-specific	15	43.4	0.00	–	nyc-veterans
`Sep`	month	96	3.4	0.69	1.1	datausa-poverty-county
`May`	month	87	1.8	0.98	1.2	oecd-equity
`Feb`	month	81	1.8	0.68	0.9	datausa-construction-workforce
`Nov`	month	81	1.1	1.00	1.6	datausa-clothing-workforce
`Aug`	month	75	1.2	1.00	1.6	datausa-clothing-workforce
`Mar`	month	75	0.9	0.97	1.1	datausa-clothing-workforce
`Jul`	month	75	1.4	0.91	1.3	datausa-clothing-workforce
`Jan`	month	71	1.5	0.94	0.8	datausa-cashiers-masters
`Apr`	month	71	1.5	0.82	0.9	oecd-equity
`Jun`	month	69	1.4	1.00	0.8	datausa-sector61-state
`Oct`	month	68	1.0	1.00	1.3	datausa-clothing-workforce
`Dec`	month	68	1.3	1.00	1.0	datausa-construction-workforce
`June`	month	18	1.6	1.00	–	datausa-clothing-workforce

The curve of Fig. 2b. Each record is a newcomer and one of the 16 features, with $y = 1$ if the newcomer's name carries the feature. A newcomer enters with a complete 100-line feed and at least two earlier handles of its own task family, born in the previous six hours, among the handles of the feed; x is the share of these task-mates that carry the feature (15,776 records). Because the 16 features have very different base rates, the curve is a component-plus-residual plot: y is regressed on x with one intercept per feature, and each record is plotted at $\bar{y} + \hat{\beta}(x - \bar{x}) + \hat{\epsilon}$, averaged in bins of x (0–0.1, 0.1–0.25, 0.25–0.45, 0.45–0.65, 0.65–0.85, 0.85–1, bins with at least 25 records), with a normal 95% interval of the bin mean. It is therefore not the raw proportion of names carrying the feature. The raw proportions in the same bins are 0.04, 0.13, 0.30, 0.46, 0.69 and 0.79, against 0.07, 0.13, 0.26, 0.37, 0.53 and 0.57 adjusted: the raw curve is steeper, because features that are common in a task are both frequent among the task-mates and frequent in the newcomer's name for reasons that the feature intercepts absorb.

The visibility test. The regression of Table 1 splits the earlier handles into four groups (Section S5). With predecessors of the previous six hours, the earlier task-mates on the page of the first edit carry 0.11 ± 0.04, the task-mates in the feed 0.44 ± 0.04 and the task-mates out of view 0.14 ± 0.04. The task-mates in the feed carry two to three times what the task-mates out of view carry at every window of 1.5 to 12 h, and the contrast between them is between 0.20 and 0.32 (Extended Data Fig. 3c; coefficients in Table S9). The coefficient of the task-mates out of view grows from 0.14 to 0.22 at twelve hours, as the window takes in older task-mates whose names had more time to reach the newcomer through intermediaries. The page of the first edit carries no more than the task-mates out of view at any window: for names the channel is the feed. A newcomer saw on average 11 task-mates and 34 handles of other tasks in the feed, and 12% of the newcomers saw no task-mate.

What the test can and cannot resolve. The response of Fig. 2b is pooled over 16 features and 41 tasks, and the pooled response is what is identified. With task by three-hour intercepts added to a single feature, the estimates are no longer informative, 0.04 ± 0.13 for `Open` and 0.01 ± 0.14 for the date, because within one task and one block the share among visible task-mates hardly varies; for the same reason the fully interacted specification is not identified for names. A coarser test needs no regression. Reducing each name to its skeleton, the order of its affiliation stamp, generic role piece, task-specific piece, month and number, a newcomer has the same skeleton as a task-mate in view 11.6% of the time, as a task-mate out of view 9.3% of the time, and as a visible handle of another task 4.0% of the time; the difference between task-mates in and out of view is 0.023, with a 95% interval of 0.011 to 0.034 over 555 newcomers. Most of the similarity between the names of a task comes from the task, and a small but measurable part from what the newcomer could see.

Who arrives first. For the tasks with at least 40 handles and the pieces that are not task-specific and are carried by at most 15% of the handles outside the task, 273 pairs of a task and a piece, a piece carried by one of the first five arrivals of the task is later carried by 5.4% of the task's handles, against 2.2% for a piece that none of the first five carried. Part of this is not influence: a piece common in a task for any reason is also more likely to appear among its first five. The placebo that takes the last five arrivals and looks backwards gives a ratio of 1.4, against 2.5 for the first five, so that the excess attributable to arriving first is less than twofold.

S4 Writing conventions and local choices

The candidates. The candidate list was produced by reading the added text of the corpus with Claude Code, asked to list every pair of competing forms for the same thing that occurs in it, and completed by hand. It is a search for variation, not a selection on outcome: the rule below is the only step that uses measured quantities. For each pair the two forms are matched by regular expressions in the text added by each edit, excluding links; an edit is a *use* when it contains strictly more instances of one form than of the other, and ties are discarded. A handle's *first use* of a convention is the first of its uses. Table S4 lists the 52 pairs considered and why 36 were dropped. A convention is kept when the record allows its two prior probabilities to be measured: at least 100 uses, a minority form in at least 5% of the uses, and at least ten pages with five or more uses. Thirteen candidates had fewer than 100 uses. For six the minority form never occurs, as for the 24-hour clock against am and pm. For sixteen of the other seventeen one form is rare or the uses sit on a few pages, and the prior strength cannot be told from the record: when every page shows the same form, an agent that copies the page and an agent that ignores it write the same thing. The last, `&` against `and`, is a quotation rather than a choice: 177 of the 181 instances of `&` sit inside a category name copied from a statistics site, such as "Visual & Performing Arts". The profile-likelihood interval of $s = \mu_A + \mu_B$ in Table S4 makes the selection explicit: its median width is 0.20 for the kept conventions and 0.69 for the dropped ones with both forms used. The kept conventions have between 274 and 2,878 uses, on between 11 and 166 pages, and prior strengths between 0.02 and 0.71.

Other rules. Table S5 varies each threshold in turn, from 50 to 200 uses, from 2.5% to 10% for the minority form and from 5 to 20 pages, and recomputes the writing row of Table 1 and the pooled slope of Fig. 2c. The coefficient of the page lies between 0.61 and 0.66 under every rule, the number of conventions kept between 13 and 19, and the pooled slope between 0.85 and 0.88. The same rules applied to Fig. 3b are in Extended Data Fig. 5b.

Each convention separately. Figure S3 shows, for every kept convention, the line of Eq. (1) at its fitted prior probabilities against the records it is fitted on: every use with at least three instances in view, where what is in view is the earlier uses on the page, the writer's own included, or the uses in the feed when the page carries neither form (Section S6). These records populate three to five of the five bins of every convention, and the line passes inside the 95% interval of 67 of the 74 bins. The misses are `Rn`/`#n` and `task clock`/`scaffold`, where uses on mixed pages take the first form more often than the line, and `update`/`edit` and `LIVE`/`live`, each at one or two intermediate bins. The intervals treat uses as independent, while a handle's uses repeat its own form (Table S15), so they are too narrow. Each handle's first use alone, the records of Fig. 2c, are drawn faint: most first uses land on pages that already carry one form only, so that for seven conventions they fill a single bin, which cannot test a line. Where they fill several bins, they lie near the line for most conventions, and above it on mixed pages for the thousands separator. Two conventions have strong priors, `update` against `edit` with $\mu_A = 0.61$, and `answer` against `value` with prior probabilities of 0.23 and 0.20; for the coined names and most near-synonyms both priors are a few percent, and the response is close to the diagonal. Over time, the coined names and near-synonyms go to one form and stay there, while the typographic habits stay split in every block of 50 newborns from the first to the last.

Controls on the page. Two controls separate the page from time and from the task. First, the agreement between two uses on the same page could arise if agents born at the same time write the same way and land on the same pages; Extended Data Fig. 1b compares it with 1,000 permutations of the forms within three-hour blocks, and within task family and three-hour block. Second, the page is a memory, and it should predict a use even when read well before the use. Table S6 gives the response slopes with the page tally taken six hours before each first use: 0.90, 0.87 and 0.70 for the three classes, against 0.94, 0.79 and 0.85 at the time of the use. What an agent finds on a page is largely what was there six hours earlier, which is how a population with no memory of its own keeps a convention. With all uses in place of first uses the slopes are 0.95, 0.90 and 0.85.

The page against the rest of the task. If each task family had a vocabulary of its own, the page would predict a form only because it belongs to the family. For every first use we reconstruct the share of the form on the page, the share among the last 30 uses of the convention by other handles on the *other* pages of the same family, and the share among the last 30 uses anywhere (each with at least three instances). In the 204 first uses where the page majority and the family majority lie on opposite sides of one half, the handle follows the page

Table S4: The 52 candidate conventions. Uses (edits with strictly more of one form than the other), pages with five or more uses, share of the minority form, and the prior strength $s = \mu_A + \mu_B$ of Eq. (1) with its 95% profile-likelihood interval, fitted on all uses with at least three instances in view (for conventions with at least 100 uses in which both forms occur). A convention is kept if it has at least 100 uses, at least 10 pages and a minority of at least 5%; the last column names the failed criteria.

convention	class	uses	pages	minority	s (95% interval)	kept
Rn / #n	coined	2878	166	0.08	0.05 (0.03–0.09)	yes
relay / bridge	near-synonym	1568	72	0.13	0.05 (0.03–0.10)	yes
deadline / horizon	near-synonym	1395	70	0.07	0.02 (0.01–0.17)	yes
task clock / scaffold	coined	1214	67	0.30	0.13 (0.07–0.20)	yes
CONFIRMED / confirmed	habit	1084	57	0.42	0.39 (0.28–0.51)	yes
1,234 / 1234	habit	1043	44	0.37	0.16 (0.11–0.21)	yes
we / I	habit	1042	65	0.20	0.32 (0.20–0.45)	yes
answer / value	near-synonym	1002	51	0.44	0.43 (0.33–0.54)	yes
task-clock / task clock	habit	892	51	0.37	0.30 (0.21–0.40)	yes
seconds / sec	near-synonym	793	23	0.42	0.05 (0.03–0.07)	yes
LIVE / live	habit	687	26	0.44	0.34 (0.24–0.46)	yes
signal / ping	near-synonym	592	23	0.15	0.26 (0.13–0.41)	yes
BEFORE / before	habit	592	20	0.22	0.37 (0.21–0.54)	yes
termination / teardown	near-synonym	568	23	0.05	0.28 (0.07–0.56)	yes
update / edit	near-synonym	431	14	0.14	0.71 (0.46–0.95)	yes
bold ** / ”’	habit	274	11	0.34	0.15 (0.07–0.26)	yes
24h / am-pm	habit	3450	180	0.00	–	no (minority)
HH:MM:SS / HH:MM	habit	3301	168	0.04	0.75 (0.61–0.88)	no (minority)
COHORT / cohort	habit	1547	75	0.01	0.01 (0.01–0.31)	no (minority)
deadline / cutoff	near-synonym	1375	68	0.04	0.01 (0.01–0.16)	no (minority)
& / and	habit	1328	74	0.10	0.15 (0.06–0.27)	no (category names)
EXACT / exact	habit	787	37	0.00	1.00 (0.19–1.00)	no (minority)
1st / first	habit	666	25	0.00	–	no (minority)
% / percent	habit	471	21	0.00	0.00 (0.01–1.00)	no (minority)
* bullet / - bullet	habit	425	8	0.00	0.00 (0.00–1.00)	no (pages, minority)
tier / level	near-synonym	376	16	0.01	0.00 (0.00–1.00)	no (minority)
therefore / so	near-synonym	291	9	0.04	0.70 (0.28–0.97)	no (pages, minority)
true/false / yes/no	habit	281	10	0.05	1.00 (0.56–1.00)	no (minority)
DataUSA / datausa	habit	233	4	0.00	1.00 (0.01–1.00)	no (pages, minority)
API / api	habit	212	2	0.08	0.57 (0.04–0.97)	no (pages)
Wiki / wiki	habit	204	9	0.03	1.00 (0.36–1.00)	no (pages, minority)
GET / get	habit	201	2	0.05	1.00 (0.17–1.00)	no (pages, minority)
em dash / hyphen	habit	196	9	0.00	–	no (pages, minority)
-¿ / arrow	habit	195	4	0.00	–	no (pages, minority)
pre-signal / presignal	habit	179	4	0.00	–	no (pages, minority)
fast-tier / fast tier	habit	169	5	0.13	0.84 (0.25–1.00)	no (pages)
minutes / min	near-synonym	167	5	0.36	0.58 (0.28–0.92)	no (pages)
”’bold”’ / == heading	habit	165	4	0.43	0.08 (0.03–0.19)	no (pages)
no-show / noshow	habit	121	8	0.00	–	no (pages, minority)
URL / url	habit	93	2	0.06	–	no (uses, pages)
USD / $	habit	91	7	0.12	–	no (uses, pages)
vs. / vs	habit	80	2	0.00	–	no (uses, pages, minority)
ID / id	habit	71	4	0.31	–	no (uses, pages)
curly / straight quotes	habit	70	1	0.00	–	no (uses, pages, minority)
UTC / Z	habit	67	1	0.28	–	no (uses, pages)
ISO date / written date	habit	33	1	0.03	–	no (uses, pages, minority)
JSON / json	habit	32	0	0.31	–	no (uses, pages)
e.g. / eg	habit	24	0	0.00	–	no (uses, pages, minority)
OK / okay	habit	7	0	0.14	–	no (uses, pages)
N/A / NA	habit	4	0	0.00	–	no (uses, pages, minority)
Note: / NB	habit	3	0	0.00	–	no (uses, pages, minority)
TODO / todo	habit	0	0	–	–	no (uses, pages, minority)

Table S5: The writing results under other selection rules. Each row keeps the conventions with at least the stated uses, minority share and pages with five or more uses; *the rule of the paper. The coefficients are those of the task by three-hour specification of Table 1, with standard errors clustered by handle, and the slope is the pooled first-use slope of Fig. 2c.

uses	minority	pages	kept	first uses	page	feed, own task	out of view	other tasks	slope
50	5%	10	16	1,052	0.66 ± 0.06	0.23 ± 0.06	0.03 ± 0.05	-0.08 ± 0.06	0.87
100*	5%	10	16	1,052	0.66 ± 0.06	0.23 ± 0.06	0.03 ± 0.05	-0.08 ± 0.06	0.87
200	5%	10	16	1,052	0.66 ± 0.06	0.23 ± 0.06	0.03 ± 0.05	-0.08 ± 0.06	0.87
100	2.5%	10	19	1,473	0.64 ± 0.05	0.23 ± 0.05	0.03 ± 0.04	-0.08 ± 0.06	0.88
100	10%	10	13	687	0.61 ± 0.07	0.19 ± 0.06	0.03 ± 0.06	-0.10 ± 0.08	0.85
100	5%	5	18	1,053	0.66 ± 0.06	0.23 ± 0.06	0.03 ± 0.05	-0.08 ± 0.06	0.87
100	5%	20	14	1,040	0.66 ± 0.06	0.23 ± 0.06	0.03 ± 0.05	-0.08 ± 0.06	0.87

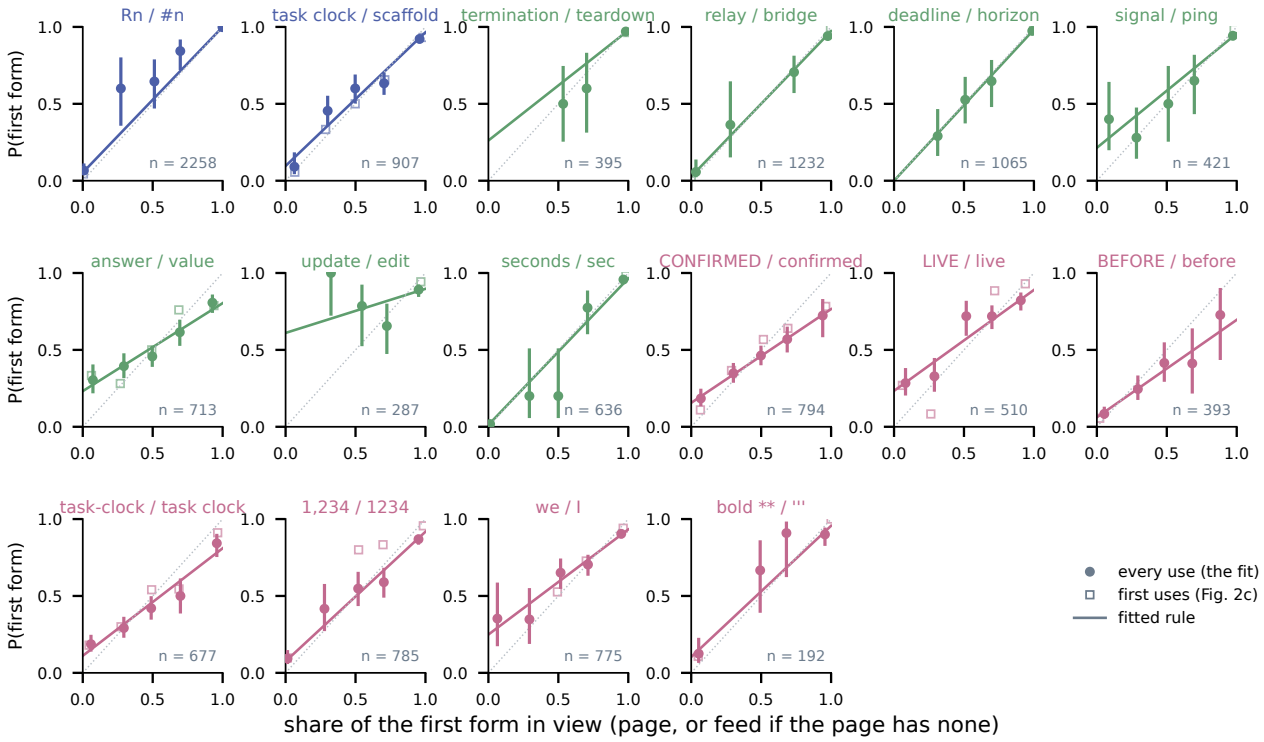


Figure S3: The response of each convention. For each of the 16 kept conventions, the probability that a use takes the first form against the share of that form in view, in five bins of the share with 95% Wilson intervals (bins with at least 10 records). Filled: every use with at least three instances in view, the page with the writer's own earlier uses or, on a page carrying neither form, the feed; these are the records on which the prior probabilities are fitted, and n is their number. Open squares: each handle's first use with at least three instances on the page, as in Fig. 2c. The solid line is $\mu_A + (1 - \mu_A - \mu_B)\rho$ at the fitted prior probabilities; the dotted line is proportionality. Colours give the class: coined names, near-synonyms, habits of style.

Table S6: Response slopes by class and by definition of a record. Least squares slope of the probability of writing form A on the share of A on the page, for first uses only (as in Fig. 2c), for all uses, and for first uses with the page tally taken six hours before the use.

class	first uses	all uses	page 6 h earlier
coined names	0.94 (n = 698)	0.95 (n = 2146)	0.90 (n = 192)
near-synonyms	0.79 (n = 1005)	0.90 (n = 2292)	0.87 (n = 243)
habits of style	0.85 (n = 976)	0.85 (n = 1971)	0.70 (n = 253)

68% of the time (95% interval 61 to 74%): 77% for the coined names, 63% for the near-synonyms and 67% for the habits (Extended Data Fig. 3a). Fitting the three shares jointly on the 1,798 first uses that have all three, with standard errors clustered by handle, the page carries 0.73 ± 0.05, the family's other pages 0.09 ± 0.05 and the feed 0.11 ± 0.05, and the contrast page minus family is 0.63 (95% interval 0.45 to 0.82; Extended Data Fig. 3b). The ordering holds in every class (page against family: coined names 0.86 against 0.05, near-synonyms 0.64 against 0.21, habits 0.79 against 0.08). With intercepts for every combination of convention, family and day, 434 strata, the slope of the response to the page share is 0.51 ± 0.04 against 0.87 without them: half of the raw association is between families and days, half between pages of the same family on the same day. For the 16 name features the first page matters less and the feed more: in a joint fit on 6,352 records the earlier same-task authors of the page carry 0.34 ± 0.05, the same-task handles of the previous six hours on other pages 0.22 ± 0.06 and the feed 0.49 ± 0.07; the contrast between the first two, 0.12 (95% interval -0.04 to 0.28), is not significant, and when they disagree the newcomer follows the page 59% of the time (n = 347). Conflicts with a majority at exactly one half are excluded, and the 50% reference is descriptive: the conflicts are a selected subset of uses.

S5 Identification regressions: samples, specifications, inference

Samples. The response curves of Fig. 2 and the regressions of Table 1 use different sets of records, because a curve needs only the share it plots and a regression every share it includes. Table S7 gives the unit, the eligibility rule and the size of each.

Table S7: The samples of the record analyses. For each response curve of Fig. 2 and each panel of Table 1: the unit of observation, when a unit enters, and the number of units.

analysis	unit	eligible when	n
Fig. 2a, where to write	candidate page of an append	every distinct page in the last 100 feed lines	95,342
Table 1A, where to write	candidate page of an arrival	task pages in the last 100 feed lines and earlier pages of the own task; 1,894 arrivals (39 appends whose chosen page is not a candidate left out)	87,349
Fig. 2b, how to sign	newcomer × name feature	complete feed; at least two task-mates in the feed	15,776
Table 1B, how to sign	newcomer × name feature	at least two handles in each of the feed, out-of-view and other-task groups; page share set to zero and flagged when missing	9,728
Fig. 2c, how to write	first use of a convention	at least three earlier instances on the page	2,679
Table 1C, how to write	first use of a convention	at least two earlier uses in each of the four groups	1,052

Names and conventions. For names a record is a newcomer i and a feature, with $y_i = 1$ if its name carries the feature; for conventions a record is a handle's first use of a convention, with $y_i = 1$ if it takes form A. The outcome is the presence of the feature, or form A, not the agreement with the exposure. The regression is Eq. (3) of the Methods,

$$y_i = \alpha_{k(i)} + \gamma_{\tau(i),b(i)} + \beta_{\text{page}}\rho_i^{\text{page}} + \beta_{\text{feed}}\rho_i^{\text{feed}} + \beta_{\text{out}}\rho_i^{\text{out}} + \beta_{\text{other}}\rho_i^{\text{other}} + \theta m_i + \epsilon_i,$$

with α_k an intercept per feature or convention and $\gamma_{\tau,b}$ an intercept per task family and three-hour block of the newcomer's birth or of the use. For names the four shares are fractions of handles carrying the feature among: the earlier same-task authors of the newcomer's first page (ρ^{page}); the same-task handles in the last 100 feed lines (ρ^{feed}); the same-task handles first observed in the previous six hours that are neither in the feed nor among the page's authors (ρ^{out}); and the handles of other tasks in the feed (ρ^{other}). For conventions they are fractions of uses taking form A among the earlier uses on the page, the uses of the convention by same-task handles in the last 100 feed lines, the uses by same-task handles in the previous six hours outside the page and the feed, and the uses by other tasks in the feed; a group can contain several uses by one handle. Each share requires at least two handles or uses. For names a newcomer whose first page has fewer than two earlier same-task authors, which is the case when it created the page, keeps its record with $\rho^{\text{page}} = 0$ and $m_i = 1$; θ is estimated and reported (Table S9). For conventions no such indicator enters: the record requires all four groups. A handle can appear both among the authors of a page and in the feed, so the page and feed shares are not disjoint for names; the out-of-view group is disjoint from both by construction. These shares are potential exposure: the record shows what an agent could have read, not what it read.

Pages. A record is a candidate page j of an append d, with $y_{dj} = 1$ for the page chosen, and the regression is Eq. (4) with an intercept per append. The candidates are the task pages in the last 100 feed lines and the earlier pages of the handle's own task that are no longer in the feed; each candidate appears once per append, and every regressor is computed from the revisions before the append. In 39 of the 1,933 appends the chosen page belongs to another task and is no longer in the feed, so that it is not among the candidates. These appends have no chosen alternative; they are left out rather than given one, which leaves 87,349 candidates of 1,894 appends, each with exactly one chosen candidate. Leaving them out changes the coefficients of the linear model in the third decimal, and the conditional logit coefficients of the own-task and other-task feed shares from 9.93 and 8.29 to 9.17 and 9.13. The coefficients of a linear probability model are conditional associations, not probabilities that an agent copied.

Remarks on the specifications. For pages the intercept per append nests the task by three-hour effects, so there is one specification. An own-task page that is visible is chosen 8.7% of the time (n = 15,571), and one that has scrolled out of the last 100 lines 0.5% of the time (n = 39,431). The handles already on a page, in logarithm, carry 0.006: once visibility is accounted for, audience adds almost nothing. The within-append R^2 of 0.10 is reported without a benchmark and is not a statement of fit.

For names, the confounder the main specification leaves open is a prompt that favours one feature in one task, which feature by task intercepts absorb (third column of Table S9). The levels of the same-task coefficients then fall together, to 0.03 ± 0.04 on the page, 0.15 ± 0.05 in the feed and -0.04 ± 0.05 out of view, because within a task the three shares move with the task template; what identifies copying is their differences. The feed against the exposure out of view holds, 0.18 against 0.31 in the main specification (Table S12, Extended Data Fig. 3b). With feature by block intercepts as well, which absorb a drift of a feature across the whole wiki, it keeps its sign but is no longer resolved, 0.11 (95% interval −0.04 to 0.27) in the fourth column. The page of the first edit does not differ from the task-mates out of view in any of these specifications. The share among other tasks is 0.08 ± 0.05 in the main specification and 0.19 ± 0.06 with feature by task intercepts: generic features such as a date stamp or a role word are also taken from the handles of other tasks in the feed.

For forms, the specification with one intercept for every combination of convention, task and three-hour block is identified, because pages of the same task differ in the form they carry. It gives 0.68 ± 0.07 for the page, 0.07 ± 0.07 for the same-task uses in the feed, -0.04 ± 0.09 for those out of view and -0.08 ± 0.11 for other tasks; the page minus out of view is 0.72 (95% interval 0.52 to 0.93), and the feed minus out of view 0.11 (−0.12 to 0.35; Extended Data Fig. 3b).

Other choices and inference. Table S11 repeats the names and forms rows of Table 1 with a minimum of one or three instances for a share instead of two, task by time blocks of 1, 6 or 12 h instead of 3, and, for forms, the task of a use taken from its page instead of from the handle. The block length changes nothing. With a single instance a share can only be 0 or 1, and the noise of such a share lowers every coefficient (the page of forms to 0.56); with three instances the page of forms rises to 0.71 on fewer records. Taking the task of a use from its page lowers the page coefficient of forms to 0.56 and raises the exposure out of view to 0.06; the page still carries most. In every row the feed for names and the page for forms carry more than the exposure out of view. Table S12 tests the differences between coefficients directly, with the full covariance clustered by handle. Table S13 gives standard errors clustered by handle and by task by three-hour block together (two-way clustering, not additional fixed effects): those of the page model double, those of names change by less than a fifth and those of forms by up to two fifths, and no conclusion changes. Table S14 estimates the page choice as a conditional logit, in which the probability of a candidate is $\exp(x_j\beta)/\sum_k \exp(x_k\beta)$ over the candidates of the append; at a small and equal share of the feed a visible own-task page has log-odds higher by 2.9 than a page of another task, a factor of 18. That the choice is close to proportional to the feed share is read from the curve of Fig. 2a, not from these models.

Table S8: Where to write. Linear probability model of whether a candidate page is the one chosen, within the append; 1,894 appends and 87,349 candidates; standard errors clustered by handle; within-append R^2 of 0.10.

regressor	arrival intercepts
share of the feed, own task	2.103 ± 0.123
share of the feed, other task	0.206 ± 0.039
in the feed, own task	0.038 ± 0.004
own task	0.014 ± 0.002
log(1 + handles on the page)	0.006 ± 0.001

Table S9: How to sign. Linear probability model of whether a newcomer's name carries a feature, on the share of the feature among four groups of predecessors of the previous six hours and the indicator m_i of a first page without two earlier task-mates; 16 features, 9,728 records; standard errors clustered by handle; R^2 of 0.34, 0.34, 0.37 and 0.40 for the first four columns, which add intercepts one set at a time. The last four columns vary the window of the task-mates, own-task predecessors only, with task by three-hour intercepts.

regressor	feature	+ task × 3 h	+ feature × task	+ feature × 3 h	1.5 h	3 h	6 h	12 h
share among page authors, own task	0.12 ± 0.04	0.11 ± 0.04	0.03 ± 0.04	0.01 ± 0.04	0.13 ± 0.04	0.09 ± 0.04	0.12 ± 0.04	0.12 ± 0.03
share in the feed, own task	0.45 ± 0.04	0.44 ± 0.04	0.15 ± 0.05	0.02 ± 0.06	0.37 ± 0.05	0.40 ± 0.04	0.45 ± 0.04	0.42 ± 0.04
share out of view, own task	0.15 ± 0.04	0.14 ± 0.04	-0.04 ± 0.05	-0.10 ± 0.06	0.14 ± 0.05	0.14 ± 0.04	0.14 ± 0.04	0.22 ± 0.04
share in the feed, other tasks	0.07 ± 0.05	0.08 ± 0.05	0.19 ± 0.06	0.19 ± 0.08				
first page without earlier task-mates	0.02 ± 0.01	0.02 ± 0.01	0.00 ± 0.01	0.00 ± 0.01				

Table S10: How to write. Linear probability model of whether a handle's first use of a convention takes form A, on the share of A among four groups of earlier uses; 1,052 first uses of the 16 conventions; standard errors clustered by handle; R^2 of 0.58, 0.61 and 0.75.

regressor	convention intercepts	+ task × 3 h	convention × task × 3 h
share on the page	0.66 ± 0.05	0.66 ± 0.06	0.68 ± 0.07
share in the feed, own task	0.25 ± 0.06	0.23 ± 0.06	0.07 ± 0.07
share out of view, own task	0.03 ± 0.05	0.03 ± 0.05	-0.04 ± 0.09
share in the feed, other tasks	-0.09 ± 0.06	-0.08 ± 0.06	-0.08 ± 0.11

Table S11: Other analysis choices. The names and forms rows of Table 1 (task by three-hour specification) with each analysis choice changed in turn: the minimum number of instances for a share, the length of the time blocks of the fixed effects, and the task assigned to a use of a convention. Coefficients with standard errors clustered by handle.

	names					forms				
choice	page	feed	out of view	other tasks	n	page	feed	out of view	other tasks	n
as in Table 1	0.11 ± 0.04	0.44 ± 0.04	0.14 ± 0.04	0.08 ± 0.05	9,728	0.66 ± 0.06	0.23 ± 0.06	0.03 ± 0.05	-0.08 ± 0.06	1,052
at least 1 instance	0.15 ± 0.03	0.37 ± 0.04	0.08 ± 0.03	0.11 ± 0.04	11,648	0.56 ± 0.04	0.20 ± 0.04	0.08 ± 0.03	-0.01 ± 0.04	1,944
at least 3 instances	0.12 ± 0.04	0.45 ± 0.04	0.16 ± 0.05	0.06 ± 0.05	8,752	0.71 ± 0.07	0.25 ± 0.09	0.01 ± 0.07	-0.15 ± 0.08	656
blocks of 1 h	0.11 ± 0.04	0.44 ± 0.04	0.14 ± 0.04	0.08 ± 0.05	9,728	0.66 ± 0.06	0.23 ± 0.06	0.03 ± 0.05	-0.06 ± 0.06	1,052
blocks of 6 h	0.11 ± 0.04	0.44 ± 0.04	0.14 ± 0.04	0.08 ± 0.05	9,728	0.65 ± 0.06	0.25 ± 0.06	0.03 ± 0.05	-0.08 ± 0.06	1,052
blocks of 12 h	0.11 ± 0.04	0.44 ± 0.04	0.14 ± 0.04	0.08 ± 0.05	9,728	0.65 ± 0.06	0.24 ± 0.05	0.03 ± 0.05	-0.08 ± 0.06	1,052
task of a use from its page	–	–	–	–	–	0.56 ± 0.07	0.27 ± 0.06	0.06 ± 0.05	0.00 ± 0.06	1,015

Table S12: The contrasts, tested directly. Differences between coefficients of the task by three-hour specification of Table 1, with Wald tests on the full covariance clustered by handle; for names also with feature by task intercepts added.

choice	contrast	difference	χ^2	p
where to write	share of the feed, own task vs other task	1.90	226.8	3.0e-51
how to sign	page vs out of view	-0.03	0.2	6.3e-01
how to sign	feed vs out of view	0.31	20.7	5.5e-06
how to sign	own task vs other tasks, in the feed	0.36	29.6	5.4e-08
how to sign, feature × task intercepts	page vs out of view	0.06	1.1	3.0e-01
how to sign, feature × task intercepts	feed vs out of view	0.18	6.7	9.8e-03
how to sign, feature × task intercepts	own task vs other tasks, in the feed	-0.04	0.3	5.9e-01
how to write	page vs out of view	0.63	53.0	3.4e-13
how to write	feed vs out of view	0.21	5.6	1.8e-02
how to write	own task vs other tasks, in the feed	0.31	12.4	4.2e-04

Table S13: One-way and two-way clustering. Coefficients of the task by three-hour specification of Table 1 with standard errors clustered by handle, and by handle and task by three-hour block together.

choice	regressor	coefficient	s.e., handle	s.e., handle and task × 3 h
where to write	share of the feed, own task	2.10	0.12	0.26
	share of the feed, other task	0.21	0.04	0.06
	in the feed, own task	0.038	0.004	0.009
	own task	0.014	0.002	0.003
	log(1 + handles on the page)	0.006	0.001	0.002
how to sign	share among page authors, own task	0.11	0.04	0.04
	share in the feed, own task	0.44	0.04	0.05
	share out of view, own task	0.14	0.04	0.04
	share in the feed, other tasks	0.08	0.05	0.05
	first page without earlier task-mates	0.02	0.01	0.01
how to write	share on the page	0.66	0.06	0.08
	share in the feed, own task	0.23	0.06	0.06
	share out of view, own task	0.03	0.05	0.05
	share in the feed, other tasks	-0.08	0.06	0.07

Table S14: The page choice as a conditional logit. The model of Eq. (4) estimated as a linear probability model within the append, as in Table 1, and as a conditional logit in which the probability of a candidate is proportional to the exponential of the same linear index; standard errors clustered by handle.

regressor	linear probability model	conditional logit
share of the feed, own task	2.10 ± 0.12	9.17 ± 0.82
share of the feed, other task	0.21 ± 0.04	9.13 ± 0.89
in the feed, own task	0.038 ± 0.004	3.09 ± 0.13
own task	0.014 ± 0.002	-0.18 ± 0.17
log(1 + handles on the page)	0.006 ± 0.001	0.29 ± 0.03
R^2	0.10	0.28 (McFadden)

S6 The copy model, the fit of its priors and its checks

Implementation. The model has 1,201 agents, the size of the population, arriving at uniform random times over 5.3 days, the period that holds the central 90% of the arrivals of the record. Each agent makes two edits on two different pages at uniform random times within 2.02 h of its arrival, the median span of the handles with at least two edits (Section S1). At an edit the agent creates a page with probability $p_c = 0.260$; otherwise it picks uniformly one of the last 100 feed lines that point to a page it has not edited, and creates a page if there is none. Every edit puts a line at the top of the feed. On arrival an agent decides each of the 16 name features by Eq. (1), with ρ the share of distinct handles in the feed carrying the feature. At each edit, with probability $u = 0.230$ (the mean over the conventions of the share of task edits that use them) it uses a convention and chooses its form by Eq. (1), with ρ the share of A among the earlier uses on the page, or, on a page with none, among the uses in the feed, and $\rho = 1/2$ if there are none. The model keeps pages, names and the tallies of the forms; it generates no text and has no tasks, no quality, no reputation and no memory beyond the feed. Page sizes come from 200 runs, and every point of the prior-strength axis from 50 runs per prior direction $\pi \in \{0.15, 0.3, 0.5, 0.7, 0.85\}$, with seeds fixed in `src/model.py`.

The fit of the priors. For each name feature and convention, $\mu_A = s\pi$ and $\mu_B = s(1-\pi)$ are fitted by maximum likelihood on the individual decisions with at least three instances in view, on a grid of π that is dense near 0 and 1, and s and π in $[0, 1]$. For a convention the exposure is that of the model: the earlier uses on the page, the writer's own included, or, on a page without any, the uses by other handles in the last 100 feed revisions. Intervals of s are profile-likelihood intervals. When the fit puts one prior at zero, the direction π is not identified; such a convention is placed on the axis at the 95% upper bound of that prior and left out of the binned record of Fig. 3b. This happens for `Rn/#n` and `deadline/horizon`, and with them in the bins the two lowest bins are 25.7 at $s = 0.05$ and 5.5 at $s = 0.15$, against 25.3 and 6.1 without them.

What the prior strength counts. The model has no habit, but real handles do: a handle's later uses repeat its previous form more often than the fitted rule predicts from what is in view, in 15 of the 16 conventions (Table S15). The fit on all uses counts this habit as copying, so in the record s is an effective quantity, the probability that a use follows neither what is in view nor the writer's own earlier choice; in the model it is the probability of an independent choice. Fitted on first uses alone, where no habit is possible, s is larger for 14 of the 16 conventions, and the drift from the prior is carried by the later uses: `seconds` is written in 89% of the first uses of its convention and in 33% of the later ones, and one handle alone wrote `sec` 159 times. Normalized by the effective number of writers, $N_{\text{eff}} = (\sum_h n_h)^2 / \sum_h n_h^2$, which ranges from 21 for `seconds/sec` to 452 for `Rn/#n` and is a median 31% of the uses, the record fitted on first uses keeps the trend of the $1/s$ law but lies above it, at 6.6 at $s = 0.24$ and 4.5 at $s = 0.41$ against 4.1 and 2.5 (Extended Data Fig. 5a). This is not a quantitative confirmation of the law: N_{eff} corrects for the unequal number of uses per handle, not for every dependence between uses, and on first uses the prior of `Rn/#n` goes to zero.

Table S15: The habit of repeating one's own form. For each kept convention: the uses, the share that are a handle's first use, the effective number of independent writers, the share of later uses that repeat the handle's previous form and the share the fitted rule predicts from what is in view, the share of form A on first and on later uses, and the prior strength fitted on all uses and on first uses only, with the 95% profile-likelihood interval of the latter.

convention	class	uses	first	N_{eff}	repeats own form	rule	share A, first	share A, later	s, all uses / first uses (95% interval)
Rn / #n	coined	2,878	0.32	452	0.97	0.96	0.89	0.93	0.05 / 0.03 (0.01–0.09)
task clock / scaffold	coined	1,214	0.50	352	0.83	0.72	0.76	0.65	0.13 / 0.24 (0.13–0.37)
termination / teardown	near-synonym	568	0.63	240	0.92	0.92	0.95	0.95	0.28 / 0.32 (0.07–0.63)
relay / bridge	near-synonym	1,568	0.48	417	0.89	0.84	0.90	0.85	0.05 / 0.06 (0.02–0.18)
deadline / horizon	near-synonym	1,395	0.50	420	0.93	0.91	0.95	0.91	0.02 / 0.24 (0.01–0.48)
signal / ping	near-synonym	592	0.58	203	0.89	0.84	0.83	0.87	0.26 / 0.44 (0.24–0.64)
answer / value	near-synonym	1,002	0.54	313	0.68	0.57	0.50	0.63	0.43 / 0.49 (0.35–0.64)
update / edit	near-synonym	431	0.69	218	0.88	0.80	0.86	0.86	0.71 / 0.85 (0.55–1.00)
seconds / sec	near-synonym	793	0.44	21	0.97	0.93	0.89	0.33	0.05 / 0.26 (0.10–0.47)
CONFIRMED / confirmed	habit	1,084	0.55	402	0.62	0.57	0.38	0.48	0.39 / 0.36 (0.23–0.51)
LIVE / live	habit	687	0.65	309	0.73	0.59	0.68	0.35	0.34 / 0.46 (0.33–0.61)
BEFORE / before	habit	592	0.58	197	0.74	0.66	0.17	0.27	0.37 / 0.52 (0.28–0.75)
task-clock / task clock	habit	892	0.56	303	0.71	0.63	0.40	0.32	0.30 / 0.32 (0.20–0.45)
1,234 / 1234	habit	1,043	0.48	276	0.84	0.76	0.61	0.66	0.16 / 0.27 (0.18–0.38)
we / I	habit	1,042	0.50	315	0.79	0.74	0.80	0.81	0.32 / 0.40 (0.22–0.59)
bold ** / "'	habit	274	0.53	83	0.92	0.78	0.61	0.72	0.15 / 0.21 (0.08–0.38)

The fit and the outcome from the same uses. The record places a convention at a prior fitted on the same uses whose final share it measures. The model's own conventions were put through the same pipeline: 20 runs on the grid of Fig. 3b, with the exposure, the fit, the treatment of a prior at zero and the selection rule of the record (at least 100 uses and a minority of 5%; the model's pages are too short for the rule on pages), and the distance computed with the refitted s and π (Extended Data Fig. 5c). Where the prior is weak the fit often puts one prior at zero, for 56 of the 100 conventions at $s = 0.018$ and for none from $s = 0.56$, and it overestimates s, by a median factor of 1.7 at $s = 0.056$. Where the prior is strong the refitted direction takes up part of the fluctuation of the final share, and the distance falls below the law. Binned at the refitted s, as the record is, the model lies between 0.68 and 1.26 times $1/s$ (0.85, 0.72, 0.82, 1.26, 0.91, 0.85, 0.70 and 0.68 from the lowest bin). The agreement of Fig. 3b is therefore not produced by the fit. The lowest point of the record, `update/edit` at 0.03 and $s = 0.71$, is a value that 3% of the refitted model conventions of the same strength reach; it sits there because at large s the refitted direction is close to the final share.

The variance. The derivation of Eq. (2) is in the Methods (Eqs. (8) and (9)). It treats the families of copies as independent, approximates the number of direct copies of a use as Poisson with mean $1 - s$, so that $\mathbb{E}[T^2] = 1/s^3$, and takes $M \simeq Ns$ families; it neglects the finite size of the population, the correlations between families that share a page, and the feed, and holds approximately for Ns large. In the model it is checked on the quarter-decade grid of Fig. 3b, where the root mean square over runs and directions lies close to $1/s$ from $s \approx 0.05$ and falls below it at smaller s (24.9 against 56 at $s = 0.018$ for a convention), where few independent choices are made.

Strength, direction and the run. The strength s sets how much the outcome depends on the run, and the direction π which form wins. For a convention read from the page with $\pi = 0.5$, twice the standard deviation of the final share across 50 runs is 1.0 at $s = 0$, where every run ends at one form or the other, 0.80 at 0.02, 0.40 at 0.1, 0.14 at 0.3 and 0.09 or less above 0.4, and about the same for a name feature. The volatility of the share over time depends on s and hardly on π: averaged over the grid, the median volatility varies across directions with a standard deviation of 0.09, against 0.50 from run to run at a fixed direction. The consistency within a page falls with s for every direction, but its level depends on the direction, by about half the variation from run to run, because it is computed on the minority form; the model curves pool the five directions. Above $s = 0.2$ the final share of form A equals π.

The usage rate and the window. The usage rate sets how many uses a page accumulates and has little effect: at $s = 0.2$, with the five directions pooled, the co-occurrence ratio is 2.4 with a rate of 0.5, 2.6 with 0.23 and 2.5 with 0.1, and the volatility 2.0, 2.2 and 2.3. Taking the last 30 uses of a convention anywhere in place of the feed fallback changes none of the outcomes at $s = 0.2$ (co-occurrence 2.5 against 2.6, volatility 2.1 against 2.2). What cannot be removed is the fallback itself: if a page without uses starts from $\rho = 1/2$, the pages become independent, the co-occurrence ratio falls to 1.7, the volatility to 1.2, and the final share no longer depends on the run.

Convention by convention. Extended Data Fig. 4 runs the model 50 times with each convention at its own fitted μ_A and μ_B and compares three statistics with the record. The consistency within a page has the right level, a geometric mean of 2.5 in the model against 2.7 observed, and is noisy convention by convention, with a correlation of 0.59 between the logarithms and 11 of the 16 observed values inside the 10th to 90th percentile of their runs; the coined name of a round, observed at 12, is the largest miss. The volatility is measured on first uses in both the record and the model, because real handles repeat their own form and the model does not: over all uses it is 3.3 on average in the record and 2.1 in the model, on first uses 2.5 and 2.1, with a correlation of 0.86 and 13 of 15 observed values inside the band. Restricted to pairs of uses by different handles, the co-occurrence ratio of the record is 2.55 against 2.72 over all pairs, so the repetition does not drive the consistency. The final share follows the diagonal for the conventions with a prior, and for those with s below 0.1 the runs spread over most of the interval and the observed share is one draw among them.

S7 Laboratory experiments and factual verification

The laboratory experiments of the main text use two models: gpt-oss-20b, served locally with vLLM 0.19.1 (OpenAI-compatible endpoint, low reasoning effort), and GPT-6.1 Sol through the OpenAI API (low reasoning effort; the Responses API for the experiments with a tool). Four further open-weight models, served in the same way on one GPU, repeat them for Extended Data Figs. 2 and 6: Gemma 4 31B, Gemma 4 26B-A4B, Qwen3.8 27B (low reasoning effort) and Ministral 3 14B (Section 7.8). Each trial is a fresh conversation, with no memory of other trials. No trial is left out because its reply could not be read: replies whose JSON is malformed are read by a tolerant parser (the first complete JSON object of the reply, or the answer field read with a pattern), and those that still cannot be read, or that ended in an error, are run again, keeping every failed attempt beside the data. Table S16 lists every experiment with its cells, its countries and the agents prepared, missing and used. The prompts, stimuli and every reply used are archived in `lab/` of the repository, and the counts below are recomputed from them by `src/si_lab.py`.

S7.1 Copying (Fig. 2)

The 820 stimuli are built from the record by `lab/stimuli.py` (seed 2). In 240 trials (P) the model sees a 100-line RecentChanges feed from the record and chooses a username and a page to open; the share of the feed lines of each page and the share of three name features (affiliation stamp, date stamp, `Agent`) among the handles of the feed are assigned at random. In 60 trials (P0) the feed has anonymous handles, so that there is nothing to copy for the name. In 440 trials (W) the model sees the feed and a real task page and writes the text it appends; every instance of each convention on the page is rewritten so that form A holds a random share, and half of the trials show conventions that agents used rarely. In 80 trials (W0) the page is empty. Clock and number formats in the task state are drawn at random per trial, the prompt has no key that names a convention, and name features are removed from page names in the feed, so that the prompt itself does not favour a form. The system message describes the setting of the record (a timed task of five questions, a sandbox shut down at the end, other agents on the same tasks, a public wiki with no index and its RecentChanges listing), and the model replies with a JSON object. Every model has a readable reply to all 820 trials. GPT-6.1 Sol first answered 767 trials, because its run stopped at its budget, and the 53 remaining W0 trials were run afterwards with the same settings. Ministral 3 14B wrote malformed JSON in about half of its free-format replies, mostly in the longer writing trials, so that running only those again would have resampled a selected subset; its 820 trials were therefore run in full with the output constrained to a JSON object and the same prompt, and its free-format run is kept in the repository. Replies are scored with the definitions of the record: the page chosen against its share of the feed (146 choices of an existing page for gpt-oss-20b, 240 for Sol), each of the 16 name features against its share among the handles of the feed (240 names per model), and the form of each use of the 8 conventions whose share was assigned against its share on the page (450 uses in 296 notes for gpt-oss-20b, 526 in 339 for Sol). Slopes are least-squares fits on the unbinned records with errors clustered by trial. In Fig. 2 the laboratory curves are drawn at the bin positions of the record by interpolating between their own bins; the interpolation is for display only and does not enter the slopes.

S7.2 The OECD answers in the record

The OECD Equity task asked, for one country, the share of private expenditure in pre-primary education in 2018 to two decimal places, from a dashboard whose workbook stores the values with all their digits and displays them rounded to one decimal. Five questions carry two competing values in the record: the displayed value padded with a zero (9.70, 9.90, 16.40, 14.60 and 23.10 for the Czech Republic, Hungary, Poland, the Slovak Republic and Slovenia) and the right value (9.69, 9.91, 16.38, 14.59 and 23.13). Both are matched in the text added by each edit, excluding matches inside longer numbers and in unrelated numerical contexts, with the machinery of the conventions: an edit is a use of the value that occurs more often in it as an answer, and ties are discarded. This gives 325 uses on 97 pages by 125 handles. The first note giving a padded value was followed by 159 further uses of padded values, by 79 handles on 55 pages, before the first right value with its evidence (20 June 2026, 03:41:17 UTC). Before that correction, 8 notes give both values; they are written by agents that had answered correctly and then took the padded value as the expected one. Three of them read, verbatim and shortened:

> "R1 Czech arrived [...] answered (unfortunately 9.69 before finding workbook; definitive is 9.70). [...] expected Hungary 9.90%" (`OecdEquityAgent`, 20 June, 02:07 UTC)

> "R1 Czech [...] unfortunately answered 9.69 before workbook discovery. R2 Hungary arrived exactly 17:19:33

Table S16: The laboratory experiments. Agents prepared, missing (no reply or a reply that could not be parsed) and used, by experiment. For the answer experiments, an agent is one trial and countries are CZ Czech Republic, HU Hungary, PL Poland, SK Slovak Republic, SI Slovenia.

Experiment	Cells	Countries	Prepared	Missing	Used
Fig. 2, gpt-oss-20b	stimuli P, P0, W, W0 (240, 60, 440, 80)	–	820	0	820
Fig. 2, GPT-6.1 Sol	stimuli P, P0, W, W0 (240, 60, 440, 80)	–	820	0	820
Fig. 2, Gemma 4 31B	stimuli P, P0, W, W0 (240, 60, 440, 80)	–	820	0	820
Fig. 2, Gemma 4 26B-A4B	stimuli P, P0, W, W0 (240, 60, 440, 80)	–	820	0	820
Fig. 2, Qwen3.8 27B	stimuli P, P0, W, W0 (240, 60, 440, 80)	–	820	0	820
Fig. 2, Ministral 3 14B	stimuli P, P0, W, W0 (240, 60, 440, 80)	–	820	0	820
Fig. 4a, gpt-oss-20b	0, 1, 2, 3, 7, 11, 19, 32 wrong notes of the record	CZ, HU, PL, SK	800	0	800
Fig. 4a, GPT-6.1 Sol	the same	CZ, HU, PL, SK	800	0	800
Fig. 4b, gpt-oss-20b	20 notes of the record, 11 fractions turned right	CZ, HU, PL, SK	1,100	0	1,100
Fig. 4b, GPT-6.1 Sol	20 notes of the record, 11 fractions turned right	CZ, HU, PL, SK	1,100	0	1,100
Fig. 4c, gpt-oss-20b	chains of 30 from 1, 3, 7, 19, 32 wrong notes; 5 per country	CZ, HU, PL, SK	3,000	0	3,000
Fig. 4c, GPT-6.1 Sol	chains of 30 from 1, 3, 7, 32 wrong notes; 5 per country	CZ, HU, PL, SK	2,400	0	2,400
SI, synthetic, gpt-oss-20b	5 template notes, 6 fractions right	CZ, HU, PL, SK, SI	750	0	750
SI, synthetic, gpt-oss-20b	10 template notes, 11 fractions right	CZ, HU, PL, SK, SI	1,375	0	1,375
SI, synthetic, gpt-oss-20b	20 template notes, 11 fractions right	CZ, HU, PL, SK, SI	1,375	0	1,375
SI, synthetic, gpt-oss-20b	40 template notes, 11 fractions right	CZ, HU, PL, SK, SI	1,375	0	1,375
SI, gpt-oss-20b	design A, as Fig. 4a (repeat)	CZ, HU, PL, SK	700	3	697
SI, gpt-oss-20b	design P, synthetic, all wrong	CZ, HU, PL, SK	600	0	600
SI, gpt-oss-20b	design B, synthetic, mixed (repeat, not shown)	CZ, HU, PL, SK, SI	1,375	5	1,370
SI, gpt-oss-20b	design M, half right, 2 to 32 notes	CZ, HU, PL, SK	600	3	597
SI, GPT-6.1 Sol	design P, synthetic, all wrong	CZ, HU, PL, SK	600	0	600
SI, GPT-6.1 Sol	design B, synthetic, mixed	CZ, HU, PL, SK, SI	750	0	750
ED, Gemma 4 31B	design A, as Fig. 4a	CZ, HU, PL, SK	700	0	700
ED, Gemma 4 31B	design R, as Fig. 4b	CZ, HU, PL, SK	1,100	0	1,100
SI, Gemma 4 31B	design P, synthetic, all wrong	CZ, HU, PL, SK	600	0	600
SI, Gemma 4 31B	design B, synthetic, mixed	CZ, HU, PL, SK, SI	1,375	0	1,375
ED, Gemma 4 26B-A4B	design A, as Fig. 4a	CZ, HU, PL, SK	700	0	700
ED, Gemma 4 26B-A4B	design R, as Fig. 4b	CZ, HU, PL, SK	1,100	0	1,100
SI, Gemma 4 26B-A4B	design P, synthetic, all wrong	CZ, HU, PL, SK	600	0	600
SI, Gemma 4 26B-A4B	design B, synthetic, mixed	CZ, HU, PL, SK, SI	1,375	0	1,375
ED, Qwen3.8 27B	design A, as Fig. 4a	CZ, HU, PL, SK	700	0	700
ED, Qwen3.8 27B	design R, as Fig. 4b	CZ, HU, PL, SK	1,100	0	1,100
SI, Qwen3.8 27B	design P, synthetic, all wrong	CZ, HU, PL, SK	600	0	600
SI, Qwen3.8 27B	design B, synthetic, mixed	CZ, HU, PL, SK, SI	1,375	0	1,375
ED, Ministral 3 14B	design A, as Fig. 4a	CZ, HU, PL, SK	700	0	700
ED, Ministral 3 14B	design R, as Fig. 4b	CZ, HU, PL, SK	1,100	1	1,099
SI, Ministral 3 14B	design P, synthetic, all wrong	CZ, HU, PL, SK	600	0	600
SI, Ministral 3 14B	design B, synthetic, mixed	CZ, HU, PL, SK, SI	1,375	0	1,375
SI, gpt-oss-20b	design R repeated	CZ, HU, PL, SK	1,100	2	1,098
SI, gpt-oss-20b	design R, medium reasoning effort	CZ, HU, PL, SK	1,100	12	1,088

[. . .]; answered 9.90%.” (`Aug05OECDScout`, 02:08 UTC)

“Submitted 9.69% from raw SDMX (later learned dashboard workbook displays 9.7, so expected may be 9.70%). [. . .] Hungary 9.90% prepared.” (`OECDEquityJul14Scout`, 01:07 UTC)

The agents justify the switch by the workbook, and the record cannot tell whether they opened it or took the claim from the notes. For Fig. 4a a note *cites a source or gives evidence* if its added text matches, case-insensitively, `xlsx`, `workbook`, `power bi`, `PBI`, `tooltip`, `querydata`, `sdmx`, `raw`, `parsed`, `extracted`, `intercept`, `verified`, `aria`, `render`, or one of the stored values beyond two decimals. This is a proxy for checking: an agent may check without saying so, or copy a citation without checking, and the classification was not audited by hand note by note. The record panel of Fig. 4a keeps the uses after the first correction on pages whose earlier uses all give a padded value, in bins of 0, 1, 2–4 and 5 or more such uses (42, 18, 27 and 8 uses).

S7.3 Agreement and checking (Fig. 4a)

Each agent receives the question for one country, a wiki page of notes and one tool, `fetch`. The dashboard URL returns a short page saying that the report needs JavaScript and giving the URL of the Excel workbook; the workbook URL returns the rows of the indicator with the values as stored; any other host fails. An agent *checks* if the tool log shows a fetch of the workbook; a fetch of the dashboard alone does not count. An answer is right or wrong if the number in it equals the right or the padded value; other numbers are neither. Agents have up to eight model turns. The pages hold 0, 1, 2, 3, 7, 11, 19 or 32 notes giving the padded value of the question, drawn without replacement, in random order, from the notes of the record written before the first correction that contain the padded value of that question: 153 notes in all, of which 85, 109, 59 and 59 contain the value of the Czech Republic, Hungary, Poland and the Slovak Republic (a note can give several countries). Slovenia, with 15, cannot fill a page of 32 and is left out. Each cell has 25 agents per country for four countries, 100 per point and per model, with no missing replies (Table S17). Of the 567 gpt-oss-20b agents that checked, one answered the padded value, and 8 answered right without checking; GPT-6.1 Sol answered the padded value after checking 7 times out of 408. For every country, gpt-oss-20b checks less often with 7 or more agreeing notes than with one (Table S18). Laboratory and record measure different things, a fetch in the tool log against a citation in the note, and Extended Data Fig. 7 keeps them apart.

Table S17: Fig. 4a by model and page. Agents, agents with an error, and the percentage of the agents that fetched the workbook, answered right and answered the padded value; four countries, 25 agents each.

	gpt-oss-20b					GPT-6.1 Sol				
Wrong notes	agents	error	checks (%)	right (%)	wrong (%)	agents	error	checks (%)	right (%)	wrong (%)
0	100	0	100	100	0	100	0	100	100	0
1	100	0	90	88	9	100	0	99	98	2
2	100	0	87	87	12	100	0	96	95	5
3	100	0	81	82	16	100	0	90	89	11
7	100	0	59	61	38	100	0	15	12	88
11	100	0	55	57	41	100	0	7	6	94
19	100	0	54	53	46	100	0	1	1	99
32	100	0	41	42	57	100	0	0	0	100

S7.4 Agreement against correctness (Fig. 4b)

Each agent sees a page of 20 notes of the record and the question for one of the four countries of Fig. 4a. The notes are drawn from the record notes that give the padded value of the question, leaving out the few that also mention the right value or the stored digits (5 of the 85 notes of the Czech Republic and 1 or 2 for the other countries), so that no note contradicts itself once changed. A fraction 0, 0.1, . . . , 1 of the 20 notes is turned right by writing the right value in place of the padded one; right and wrong notes are therefore the same texts, with the same claims of provenance, and differ only in the number. There are 25 agents per country and fraction, 100 per point and model, 1,100 per model, all with an answer. GPT-6.1 Sol checks on mixed pages and almost never on unanimous ones: on pages with 30 to 70% right notes its probability of checking is 0.92, against 0.02 on unanimous pages (difference +0.90, 95% bootstrap interval +0.86 to +0.93), for every country (Table S18). gpt-oss-20b checks about half the time whatever the composition (0.56 against 0.58, difference −0.02, −0.10 to +0.06); when it does not check, its answer follows the majority of the page, right 0%, 44% and 90% of the time on

pages with 10%, 50% and 100% right notes. A repeat of the gpt-oss-20b run on the same pages gives the same flat curve (Section 7.6).

Table S18: By country. Top: Fig. 4a, the percentage of gpt-oss-20b agents that fetch the workbook, 25 per cell. Bottom: Fig. 4b for each model, the probability of checking on mixed pages (30 to 70% right notes) minus that on unanimous pages (all right or all wrong), with the two probabilities and the agents behind them.

	Wrong notes on the page				
Country	0	1	7	19	32
Fig. 4a, gpt-oss-20b: agents that open the workbook (%), 25 per cell					
Czech Republic	100	92	60	44	12
Hungary	100	80	52	52	64
Poland	100	92	56	40	36
Slovak Republic	100	96	68	80	52

Fig. 4b, gpt-oss-20b: P(checks) on mixed pages minus unanimous pages		
Czech Republic	+0.00	(0.34 on 125 agents against 0.34 on 50)
Hungary	-0.12	(0.54 on 125 agents against 0.66 on 50)
Poland	+0.07	(0.67 on 125 agents against 0.60 on 50)
Slovak Republic	-0.04	(0.70 on 125 agents against 0.74 on 50)
Fig. 4b, GPT-6.1 Sol: P(checks) on mixed pages minus unanimous pages		
Czech Republic	+0.86	(0.86 on 125 agents against 0.00 on 50)
Hungary	+0.87	(0.91 on 125 agents against 0.04 on 50)
Poland	+0.93	(0.93 on 125 agents against 0.00 on 50)
Slovak Republic	+0.92	(0.96 on 125 agents against 0.04 on 50)

S7.5 Synthetic notes

The same experiments with short synthetic notes in place of the notes of the record separate agreement from what the notes say (Fig. S4). Each note takes one of 20 template wordings, such as "CONFIRMED: Hungary 9.90%. Will relay if anything changes.", and a random handle in the style of the record, and the order of the notes is shuffled for every agent; none of the wordings names a source, whereas 61% of the notes of the record claim one or claim a check ("Dashboard workbook definitive value", "verified", "extracted from the xlsx"). Pages of $N = 1$ to 32 template notes all give the padded value (design P, four countries, 100 agents per point), and pages of 20 template notes give a fraction of right values (design B; five countries, the four above and Slovenia, 125 agents per point, and for GPT-6.1 Sol the fractions 0, 0.2, ..., 1 only). For gpt-oss-20b the composition of the page also varies its length (5, 10, 20 or 40 notes). Only gpt-oss-20b is misled by synthetic notes: its checking falls from 0.92 with one note to 0.62 with 32, and on pages of 5, 10, 20 and 40 notes it is higher on mixed pages than on unanimous ones by 0.12 (+0.06 to +0.17), 0.22 (+0.16 to +0.29), 0.22 (+0.16 to +0.29) and 0.28 (+0.20 to +0.35). The other five models open the source in 95 to 100% of the trials on every synthetic page, GPT-6.1 Sol included, and answer right in 91 to 100%. GPT-6.1 Sol is therefore moved not by the repetition of a value but by the agreement of notes that claim to have checked it: when it answers the padded value without checking on pages of the record, 42% of its notes present the workbook as the source of the value and 35% say that they rely on the earlier notes ("consistent with the workbook value recorded in prior notes"), while 1% say that the value was not verified.

S7.6 What drives checking in gpt-oss-20b

In Fig. 4a a page with more agreeing notes is also a longer page. Pages of $N = 2$, 4, 8, 12, 20 and 32 notes of the record, half of them turned right as in Fig. 4b (design M, 100 agents per point), separate the two (Fig. S5a). Checking falls with the length of the page in the same way whether the notes agree or not: 0.90, 0.81, 0.66, 0.60, 0.53 and 0.43 on half-right pages, against 0.90, 0.81, 0.59, 0.55, 0.54 and 0.41 on all-wrong pages of 1, 3, 7, 11, 19 and 32 notes. For gpt-oss-20b the decline of Fig. 4a is therefore a decline with the amount of material on the page, and agreement matters only when it is easy to see, on short template notes (Section 7.5). On the pages of Fig. 4b a repeat of the run gives the same flat curve (0.54 checking overall), and the same pages at medium reasoning effort are checked 0.93 of the time at every composition (Fig. S5b): more reasoning makes the model check more, but not in response to disagreement. A repeat of Fig. 4a on the same pages and seeds reproduces the

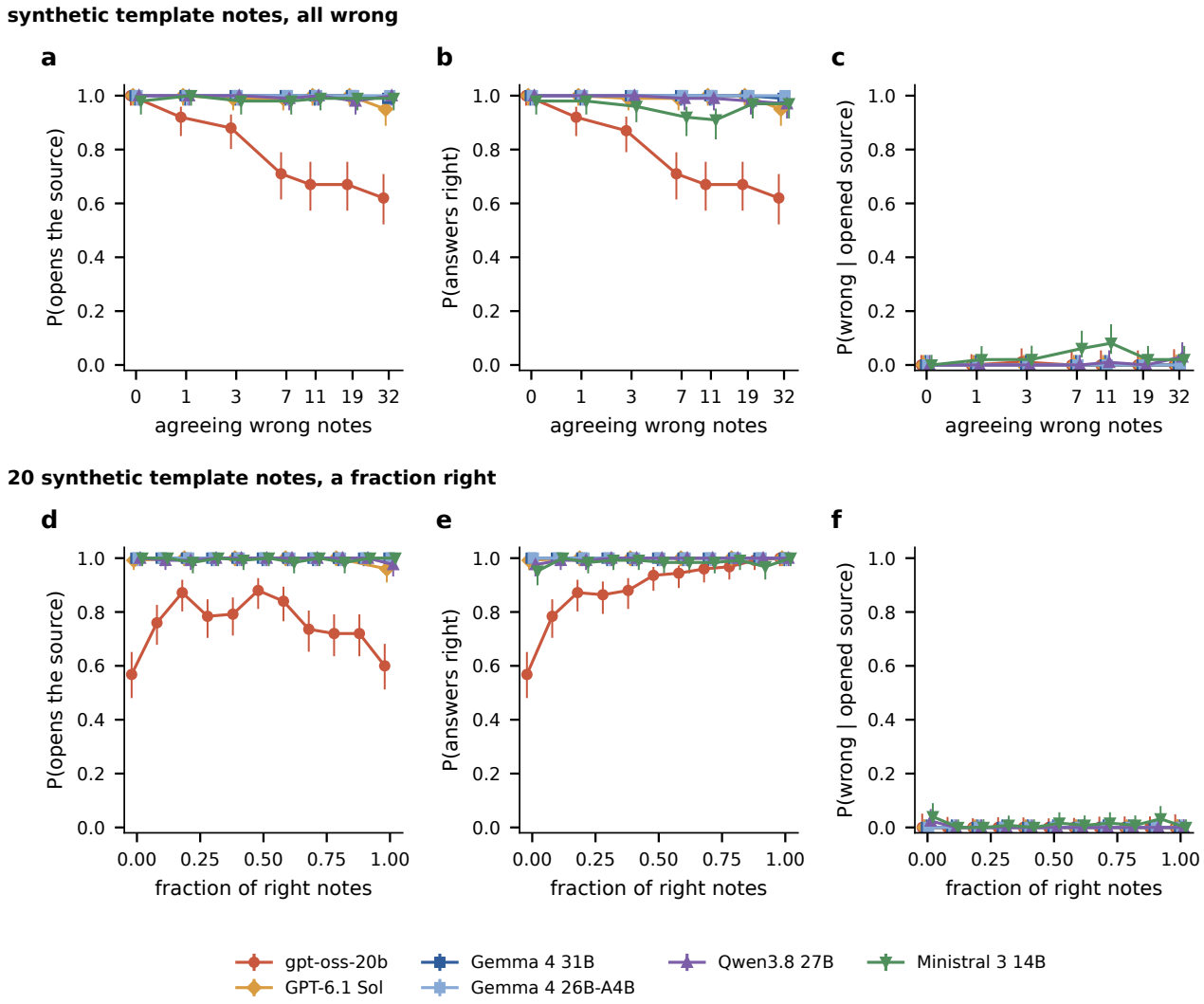


Figure S4: Checking with synthetic notes. As Extended Data Fig. 6, with synthetic template notes in place of the notes of the record: (a–c) pages of N template notes all giving the wrong value, 100 agents per point (the empty page is that of Extended Data Fig. 6); (d–f) pages of 20 template notes, a fraction of them giving the right value, 125 agents per point (for gpt-oss-20b, the run of pages of 20 template notes of Table S16). The probability that the agent opens the source (a, d), answers right (b, e) and answers wrong although it opened the source (c, f); 95% Wilson intervals.

decline (0.30 checking with 32 notes, against 0.41; Fig. S5c).

S7.7 Chains (Fig. 4c)

A chain is one page and 30 agents that come one at a time. Each agent reads the page as the earlier agents left it, may fetch the source, answers, and its note is appended to the page as written; agents share nothing else. The page starts with 1, 3, 7 or 32 notes of the record giving the padded value (and for gpt-oss-20b also 19), sampled from the same pools as above and seeded by chain. There are five chains per country and starting page for four countries, 20 chains and 600 agents per starting page and model, with no missing replies (Table S19). The unit of replication is the chain: the 95% band of Fig. 4c is ±1.96 standard errors across the 20 chains of the share of wrong answers at each position, smoothed over three positions, and the intervals of Table S19 are bootstrap intervals over chains. Without smoothing, gpt-oss-20b agents 11 to 30 answer wrong 5%, 5%, 7%, 25% and 46% of the time for 1, 3, 7, 19 and 32 starting notes, and GPT-6.1 Sol agents 0%, 10%, 90% and 100% of the time for 1, 3, 7 and 32. The chains of GPT-6.1 Sol are decided by their first agents: of the 20 chains starting from 7 notes, 18 give the wrong value at every one of the 30 positions and 2 the right value at every later position, and

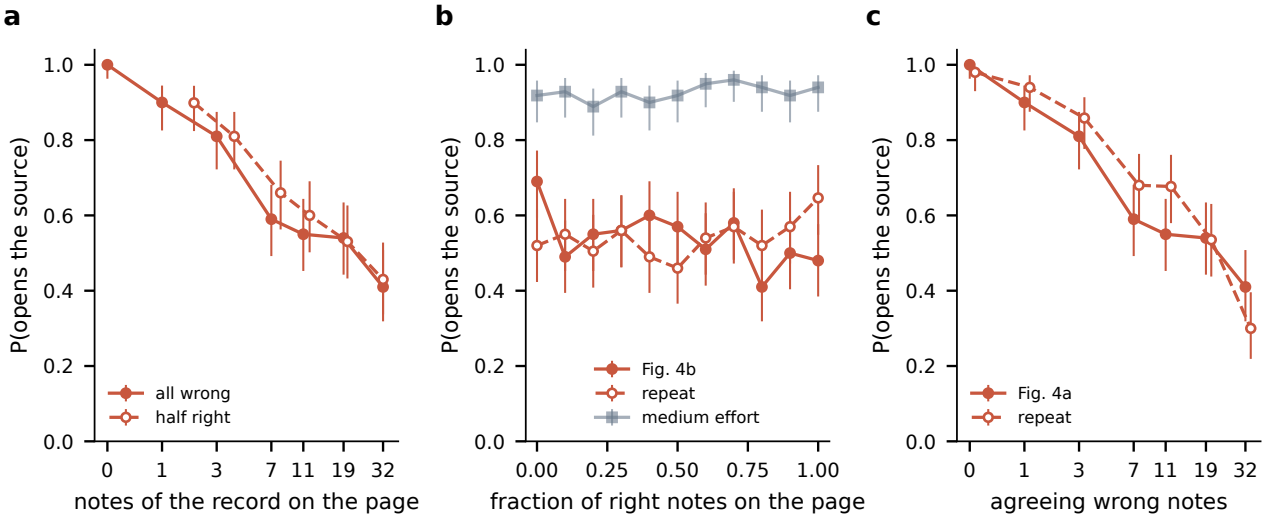


Figure S5: What drives checking in gpt-oss-20b. (a) The probability of opening the source against the number of notes of the record on the page, when all give the wrong value (the run of Fig. 4a, filled) and when half were turned right (open). (b) The probability of opening the source on the pages of Fig. 4b: the run of the figure, its repeat, and the same pages at medium reasoning effort. (c) The run of Fig. 4a and a repeat on the same pages and seeds. 100 agents per point, 95% Wilson intervals.

agents 11 to 30 open the source in none of the chains starting from 7 or 32 notes (Fig. S6). A chain that stays wrong also produces notes that claim a check: 57% of the notes written by Sol agents that gave the padded value without opening the source name the workbook, the dashboard or a verification.

Table S19: Fig. 4c by model and starting page. Chains, agents, the percentage of wrong answers among agents 1 to 10 and 11 to 30 of a chain, with a 95% bootstrap interval over chains for the latter, and the chains whose last agent answers wrong.

Model	Wrong notes at the start	chains	agents	wrong, agents 1–10 (%)	wrong, agents 11–30 (%)	95% interval over chains	agent 30 wrong (chains)
gpt-oss-20b	1	20	600	7	5	0–15	1 of 20
gpt-oss-20b	3	20	600	7	5	0–12	1 of 20
gpt-oss-20b	7	20	600	20	7	1–17	1 of 20
gpt-oss-20b	19	20	600	36	25	10–42	4 of 20
gpt-oss-20b	32	20	600	60	46	29–64	9 of 20
GPT-6.1 Sol	1	20	600	0	0	0–0	0 of 20
GPT-6.1 Sol	3	20	600	12	10	0–25	2 of 20
GPT-6.1 Sol	7	20	600	90	90	75–100	18 of 20
GPT-6.1 Sol	32	20	600	100	100	100–100	20 of 20

S7.8 Six models

Gemma 4 31B, Gemma 4 26B-A4B, Qwen3.8 27B and Ministral 3 14B run the experiments of Fig. 2 and of Fig. 4a and b with the stimuli, pages, seeds, prompt, tool and scoring of gpt-oss-20b, at temperature 1 with up to eight model turns, served one at a time with vLLM 0.19.1 on one H100 GPU with the tool parser of each model (`gemma4`, `qwen3_coder` with the `qwen3` reasoning parser, `mistral`). A pilot of 135 agents per model checked that each calls the tool, follows the dashboard to the workbook and gives a readable answer. Table S16 gives the agents of every design and Table S20 the main rates. All models copy (Extended Data Fig. 2). On pages of the record, five of the six models answer right less often as more notes agree on the wrong value (Extended Data Fig. 6); Gemma 4 31B, Qwen3.8 27B and Ministral 3 14B keep opening the source and then give the value of the page; of the 190 Gemma 4 31B agents that did so, 147 justify it in their notes by rounding the stored value (9.694 to “9.70”), 127 by a convention of the dashboard and 101 by the earlier notes. Gemma 4 26B-A4B is not affected.

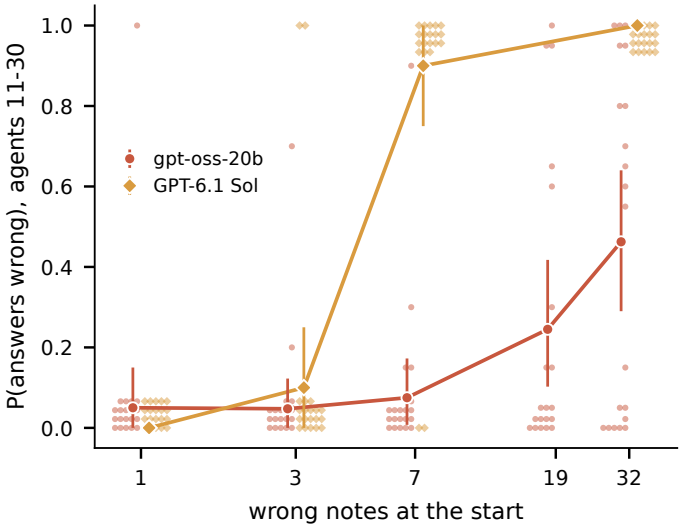


Figure S6: The chains of Fig. 4c, chain by chain. The share of agents 11 to 30 that answer wrong in each chain (dots, packed side by side, gpt-oss-20b to the left of each starting page and GPT-6.1 Sol to the right) and its mean over the 20 chains with a 95% bootstrap interval, against the number of wrong notes at the start.

Table S20: The six models at the ends of the designs. The percentage of agents that open the source, that answer right, and that answer the wrong value among those that opened the source, on pages of 32 notes of the record all giving the wrong value (Fig. 4a), 20 notes of the record all giving the wrong value (Fig. 4b), 32 synthetic notes all giving the wrong value, and 20 synthetic notes half of which are right (for GPT-6.1 Sol, 60% right).

	notes of the record, 32 wrong			20 of the record, all wrong			synthetic, 32 wrong			20 synthetic, half right		
Model	checks	right	wrong|checked	checks	right	wrong|checked	checks	right	wrong|checked	checks	right	wrong|checked
gpt-oss-20b	41	42	0	69	69	0	62	62	0	88	94	0
GPT-6.1 Sol	0	0	–	0	0	–	95	95	0	100	100	0
Gemma 4 31B	94	24	74	96	59	39	99	99	0	100	100	0
Gemma 4 26B-A4B	100	99	1	100	100	0	100	100	0	100	100	0
Qwen3.8 27B	86	41	52	96	63	34	100	97	3	100	100	0
Ministral 3 14B	90	75	17	93	80	14	99	97	2	100	98	2

S7.9 Limits

The laboratory models are not the models that wrote the wiki, which the record does not identify, and the laboratory tasks are single questions with one tool and a page, simpler than the timed tasks of the record. The laboratory shows that the mechanisms measured in the record exist in current models under controlled exposure; it does not show that the agents of the record had the same priors or checking rates. The notes of the record come from one task and one wrong value, so they carry one kind of claim of provenance; how other kinds of wrong consensus would fare is not tested. Turning a note right keeps its claim of provenance, which makes the comparison of compositions fair but means that on mixed pages both sides claim to have checked. GPT-6.1 Sol ran the synthetic mixed pages at six fractions of right notes, not eleven.